

# Comparing Random Forest, Support Vector, and K-Nearest Neighbors Classifiers on Sentinel-2 Satellite Imagery

**Group Assignment Report**  
(Machine Learning)

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# Introduction

Satellite imagery has revolutionized the way we observe and analyze Earth's surface, enabling detailed monitoring of land cover, environmental changes, and human activities. The Sentinel-2 satellite, with its multi-spectral imaging capabilities, provides moderate-resolution data across various spectral bands, making it a valuable resource for remote sensing and land cover classification tasks.

In this project, researchers leverage the EuroSAT dataset, which consists of Sentinel-2 image tiles labeled by land cover type, to compare the performance of three popular machine learning classifiers: Random Forest (RF), Support Vector Machine (SVM), and k-Nearest Neighbors (kNN). Our objective is to evaluate how well each model can classify land cover types based on selected spectral bands (blue, green, red, and near-infrared), and to analyze their strengths and weaknesses in terms of accuracy, generalization, and robustness to class imbalance.

Through systematic data processing, feature analysis, and model optimization, we aim to provide insights into the suitability of these classifiers for satellite image classification and to identify the best approach for extracting meaningful information from remote sensing data.

## Dataset Overview

The research utilizes the EuroSAT dataset, a large-scale collection of Sentinel-2 satellite imagery designed for land use and land cover classification. The dataset includes 10 classes of land cover: AnnualCrop, Forest, HerbaceousVegetation, Highway, Industrial, Pasture, PermanentCrop, Residential, River, and SeaLake. It is comprised of thousands of 64x64 pixel image tiles, each containing 13 spectral bands captured at a 10-meter ground sampling distance (GSD). This project focuses on four key spectral bands: blue, green, red, and near-infrared (NIR). These bands are the most relevant for distinguishing terrestrial features and vegetation.

## Dataset Source

- EuroSAT Dataset: [Kaggle - EuroSAT Dataset]  
(<https://www.kaggle.com/datasets/apollo2506/eurosat-dataset?resource=download-directory&select=EuroSATallBands>)
- Sentinel-2 Bands Reference: [Sentinel Hub - Band Information]  
(<https://custom-scripts.sentinel-hub.com/custom-scripts/sentinel-2/bands/>)

## Literature Review

The article *Comparison of Random Forest, k-Nearest Neighbor, and Support Vector Machine Classifiers for Land Cover Classification Using Sentinel-2 Imagery* by Phan Thanh Noi and Martin Kappas (2018) investigates the effectiveness of three popular machine learning

algorithms—Random Forest Classifier (RFC), k-Nearest Neighbor (kNN), and Support Vector Classifier (SVC)—in classifying land cover using Sentinel-2 satellite imagery. The data was collected over a 30 by 30 kilometer-squared area in Vietnam, where 14 datasets were tested using different training sample sizes and varying levels of class balance. The results showed consistently high testing accuracy across all classifiers, ranging from 90% to 95%. SVC emerged as the most robust, maintaining high accuracy regardless of training sample size, while RFC and kNN showed greater sensitivity to sample size variations. When training samples exceeded 750 pixels per class—representing about 0.25% of the study area—all classifiers achieved similarly high testing accuracy (above 93.85%). The study concludes that adequate sample size is critical for optimal classification performance, especially when using imbalanced datasets (Phan Thanh Noi & Kappas, 2018).

## Objective

The primary objective of this project is to evaluate and compare the performance of three machine learning classifiers—Random Forest, Support Vector Machine, and k-Nearest Neighbors—on land cover classification using Sentinel-2 satellite imagery from the EuroSAT dataset. By leveraging four key spectral bands, this project will:

- Develop optimized models for each classifier using hyperparameter tuning.
- Assess model performance using robust metrics such as accuracy, precision, recall, F1-score, and ROC AUC.
- Analyze the strengths and limitations of each approach in handling class imbalance and complex remote sensing data.
- Provide insights into which classifier is best suited for Sentinel-2 satellite land cover classification and discuss the underlying reasons for their performance differences.

## Data Preparation

Given the original dataset's size – approximately 2.8 GB, over 113 million pixels for the selected bands – direct processing is computationally intensive. To enable efficient model development and evaluation, the following steps were performed:

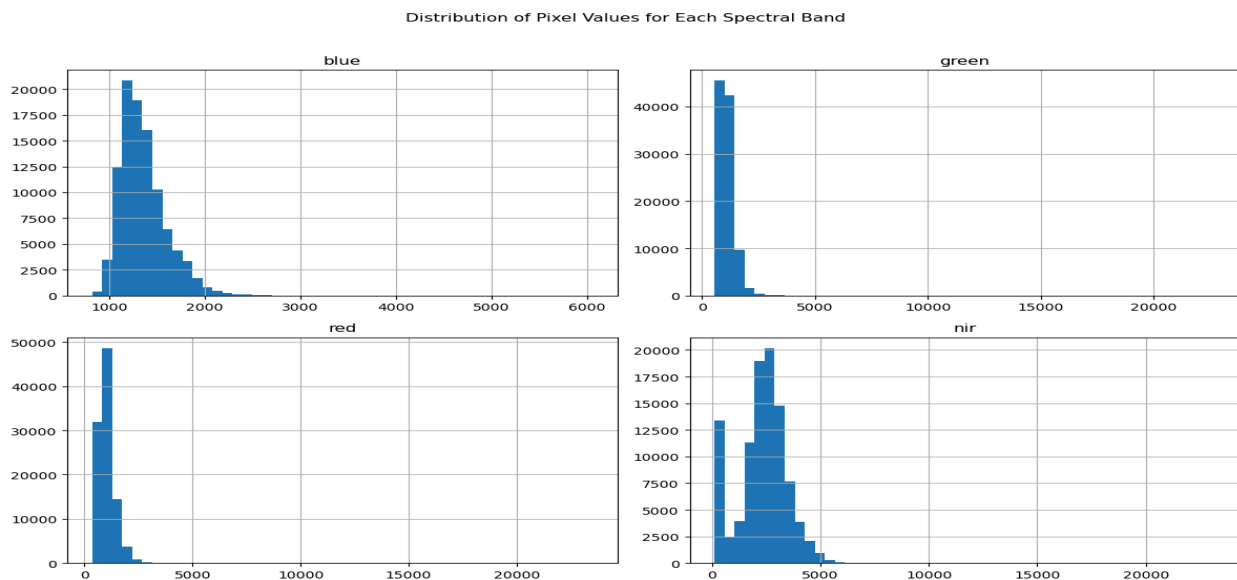
- Band Selection: extracted only the blue, green, red, and NIR bands from each image tile.
- Stratified Sampling: reduced the dataset to a representative subset of 100,000 pixels, ensuring class proportions are preserved using stratified sampling.
- Data Storage: saved the processed subset as a compressed NumPy archive (`.npz`), containing feature matrix `X`, target vector `y`, and class label list.
- Features: four spectral bands per pixel (blue, green, red, NIR), all numerical.
- Target: multi-class land cover labels (10 classes).
- Class Imbalance: some classes are more prevalent than others, requiring stratification during train-test split and careful evaluation using metrics beyond accuracy.

- Scikit-learn Intel Extension Accelerator: This AI-powered extension offers significant performance improvements for a range of Scikit-learn algorithms and applications (Intel Corporation, 2025).

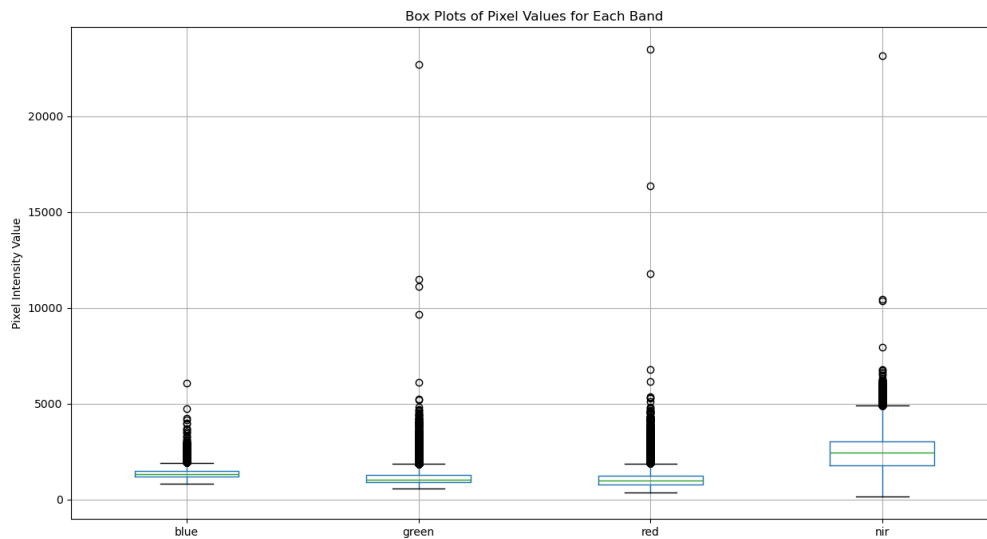
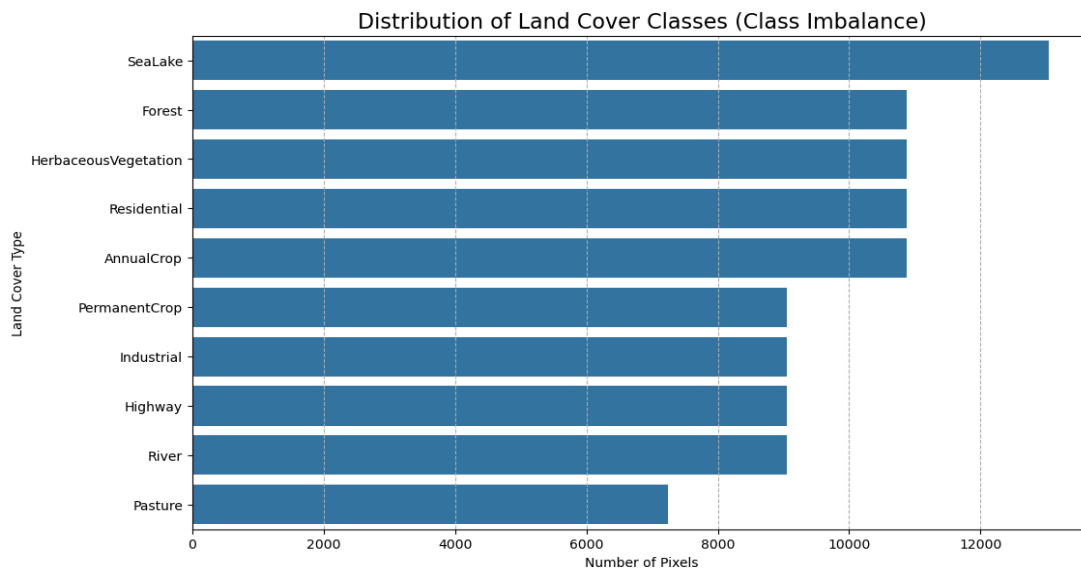
This curated dataset provides a robust foundation for comparing machine learning classifiers selected for evaluation, enabling both efficient computation and meaningful evaluation.

## Data Exploration

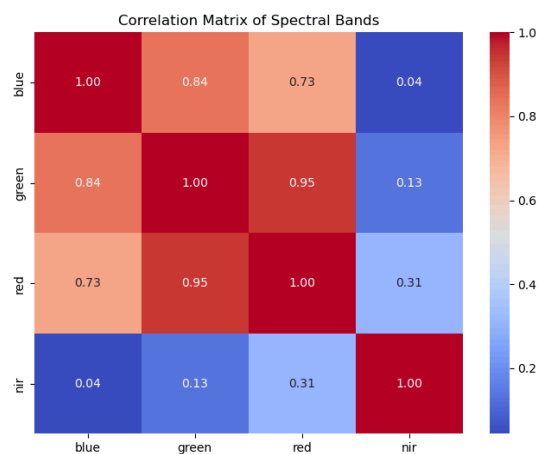
The histogram plot illustrates the distribution of pixel intensity values for the four spectral bands in the EuroSAT dataset subset. The visible bands (blue, green, red) display similar unimodal distributions with most values concentrated in the lower range, while the NIR band exhibits a broader and more variable spread. The observed differences in scale and distribution among the bands suggest that feature scaling will be important for downstream machine learning models, especially those sensitive to feature magnitude, such as SVC and kNN.



The bar plot shows that the EuroSAT dataset subset has a significant class imbalance, with some land cover types, such as Forest and SeaLake, being much more prevalent than others, like Pasture and Highway.



The above plot illustrates that the visible spectral bands (blue, green, red) are strongly correlated with each other, while the non-visible (NIR) band is less correlated, providing complementary information for land cover classification. This is also notable in the heatmap below.



# Model Design and Evaluation

## Data Stratification Reduction

To address the computational challenges posed by the large EuroSAT dataset, stratified sampling was employed to reduce the data from over 113 million pixels to a manageable subset of 100,000 pixels. This approach ensured that the proportion of each land cover class was preserved, maintaining representative class distributions and mitigating class imbalance. By selecting pixels proportionally from each class and saving the result as a compressed NumPy archive, the process enables efficient data loading and supports robust model development while preserving the diversity and structure of the original dataset.

## Train-Test-Split

A stratified train-test split was applied to the subset to ensure proper proportions of each class in  $y$  are preserved. This is critically important due to the class imbalance notable by the number of image files, and therefore pixels, differing per class. This imbalance is shown in the raw data (before subsetting) read in the cell's outputs. For example, AnnualCrop has 3000 images, SeaLake has 3597 images, but Pasture only has 2000 images.

```
#Train-Test Split dataset with stratification to ensure proper proportion of each
class in y is preserved
X_train, X_test, y_train, y_test = train_test_split(X, y, stratify=y, test_size=0.2,
random_state=12)
```

## Random Forest Classifier (RFC)

Random Forest is an ensemble learning method that constructs multiple decision trees during training and outputs the mode of their predictions for classification tasks. Each tree is trained on a random subset of the data and features, which helps reduce overfitting and improves generalization. Its ability to capture non-linear relationships and feature interactions makes it well-suited for remote sensing and image classification tasks.

### RFC Model A Results

```
{'n_estimators': 250, 'min_samples_split': 10, 'max_features': 1, 'max_depth': None}
Optimized Random Forest Classification Training Accuracy: 86.85%
Optimized Random Forest Classification Testing Accuracy: 56.32%
```

### RFC Model B Results

```
{'n_estimators': 100, 'min_samples_split': 5, 'max_features': 4, 'max_depth': 10}
Optimized Random Forest Classification Training Accuracy: 55.68%
Optimized Random Forest Classification Testing Accuracy: 53.55%
```

## RFC Models Overfitting and Accuracy Comparison

Based on the high training accuracy and significantly lower testing (overall) accuracy, it can be stated with certainty that the Random Forest Classification (RFC) Model A severely overfitted. The similar training and testing (overall) accuracy showcased by RFC Model B indicates good generalization with minimal overfitting. The reason RFC Model B generalized significantly better comes down to the maximum tree depth allowed during the tuning of the model. Based on the assignments and some testing, it appears that if a RFC model is provided with the option to extend its tree depth as far as possible (i.e. `max_depth=None`), it will gladly do so to maximize the scoring parameter.

While RFC Model A does have better training and testing accuracy, there is a high likelihood that it would fail or, at the very least, have very low testing accuracy on new unseen data, especially if any distribution shifts occur, such as seasonal or lighting differences within the Sentinel-2 imagery. RFC Model B, on the other hand, may have a slightly lower testing score, but it generalizes much better with minimal difference in the ROC AUC curves, resulting in a more reliable model.

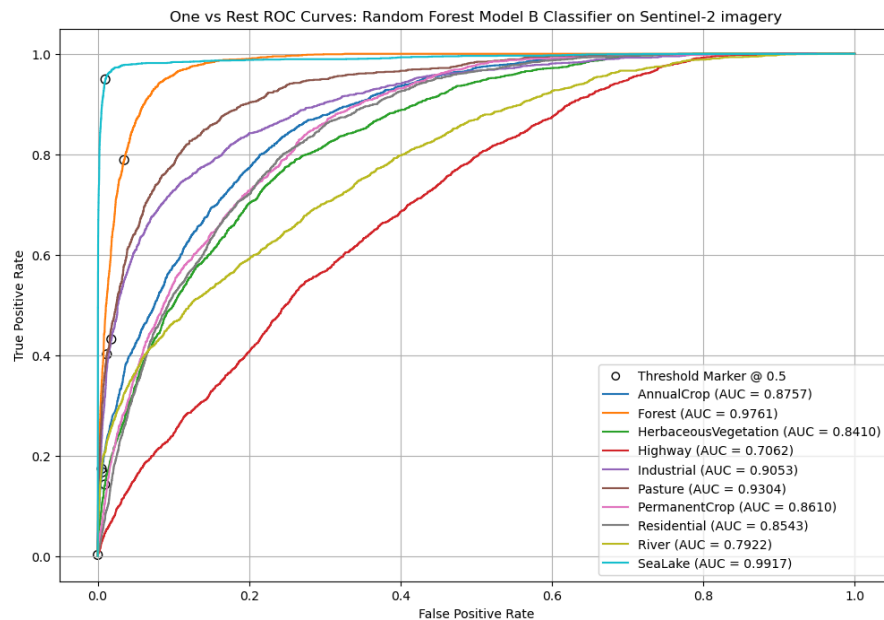
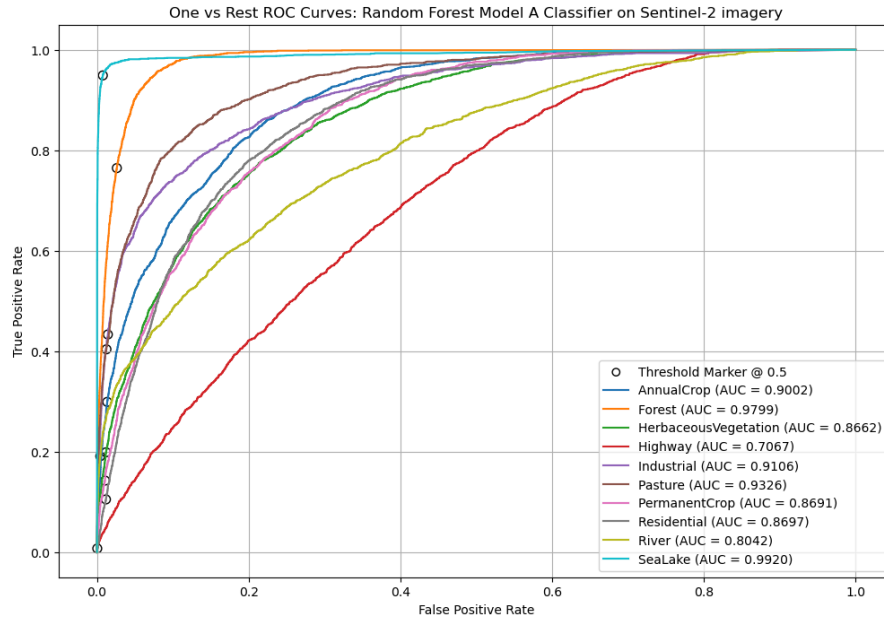
## RFC Models ROC and AUC Curves Comparison

Looking at the ROC curves with thresholds set to 0.5 (default) per class, it is notable that the ROC AUC value dropped one percentage point between Model A and Model B, respectively. All of the AUC curve values also dropped slightly. Some, like Annual Crop, dropped more than others, like SeaLake. The drop in ROC AUC (OvsR) value can be explained by the decrease in testing accuracy, as can the decline in the AUC curve.

Most of the 0.5 thresholds for both models are near the far-lower-left part of the curves (near 0). This suggests that the models are not well-calibrated, which may lead to discrepancies between their accuracy scores and the AUC, as observed in the results. However, most of the curves rise quickly with minimal rightward movement, along with the far-left threshold position, indicating that the models are capable of, and, in most cases, provide low false positive rates. These results indicate that precision falls within acceptable limits. Prior to deploying any of these models in a production setting, it would be necessary to adjust the classification threshold for each class to optimize both recall and precision, ideally approaching the upper-left corner of the precision-recall space.

While some classes, such as SeaLake and Forest, show near-perfect ROC curves and separability, others, such as Highway, have a fanned-out curve, indicating potential overlap with other classes in feature space. For example, trees or pastureland overlapping or near a highway may contribute to a fanned-out curve. Another potential and correlated reason why some classes performed much better than others in both models is due to the size of the features and the 10-meter resolution of the imagery. Smaller or narrower land features, such as a highway or river, may only cover 1 to 2 pixels in width, making them less prevalent in the imagery and more likely to be contained in 'mixed pixels' rather than 'pure pixels', diluting the clarity of the pixel belonging to a given class in feature space. Larger and easily differentiable land features, such

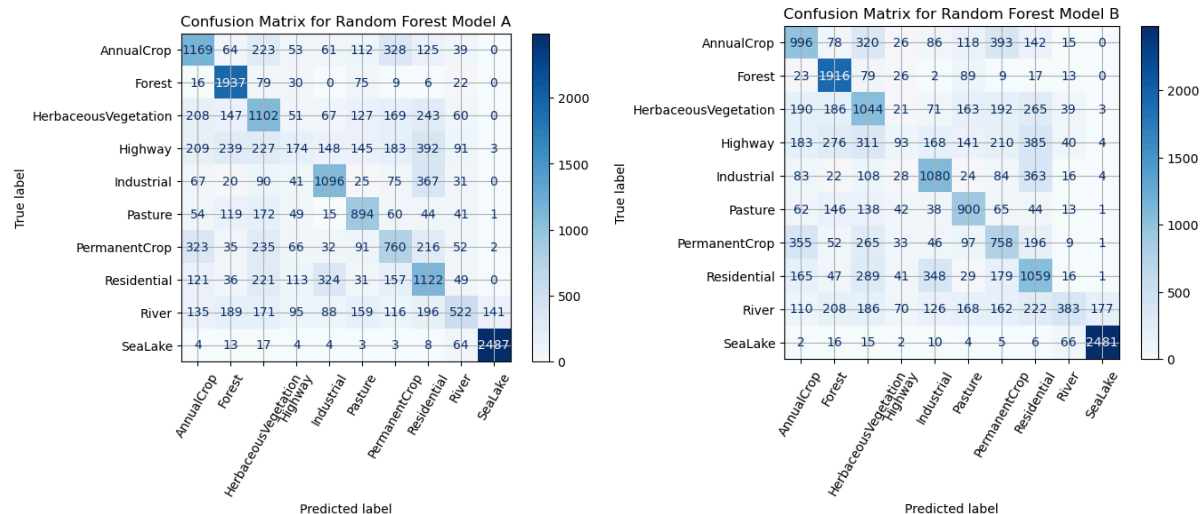
as AnnualCrop and Forest would inherently do better due to their discernable features in feature space, while land cover types, such as HerbaceousVegetation and Pastureland are more similar in RGBN pixels, making them less discernible in feature space. This easily explains why Highway and River classes struggled so much. Textural modeling would need to be implemented to help with such discernments. Rivers tend to vary quite a bit in colour and in particulate matter, leading to large variations in RGBN band values. This large variation makes rivers, and sometimes lakes, quite difficult to classify, and explains why both models struggled with this class.



## RFC Models Confusion Matrices Comparison

Confusion Matrix Accuracy % for Model A = 56.32%

Confusion Matrix Accuracy % for Model B = 53.55%



The above confusion matrix accuracy clearly shows that Model A slightly outperformed with an accuracy of 56.32%, while Model B had around 53.55%. Model A performed well in predicting classes like Forest, HerbaceousVegetation and SeaLake with high positive counts. It struggled with differentiating between classes like Residential vs River, and AnnualCorp vs Pasture, however. Model B showed clear class discrimination for some classes, but frequent misclassifications between similar classes, such as AnnnualCrop vs PermanentCorp, and River vs Residential.

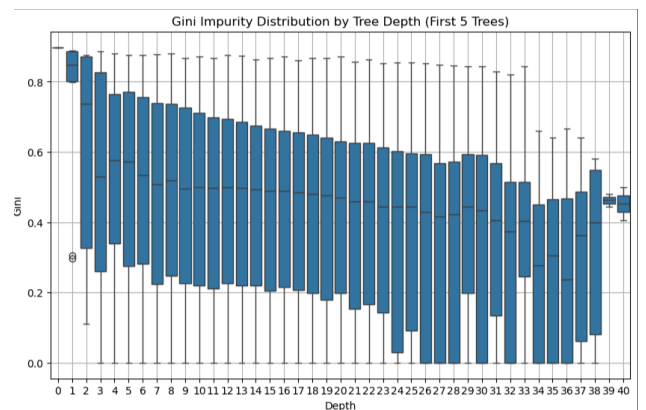
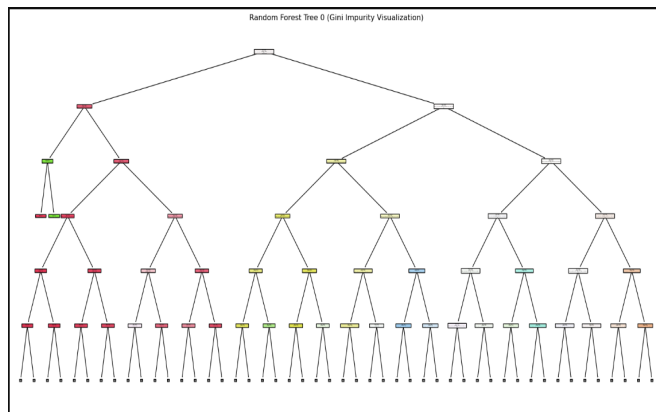
Both of the models have shown strong performance in predicting the SeaLake and Forest classes with minimal errors. Both models also showed the Highway class as the most confused class, but more so in Model B.

## RFC Models F1-Score, Recall, and Precision Results and Comparison

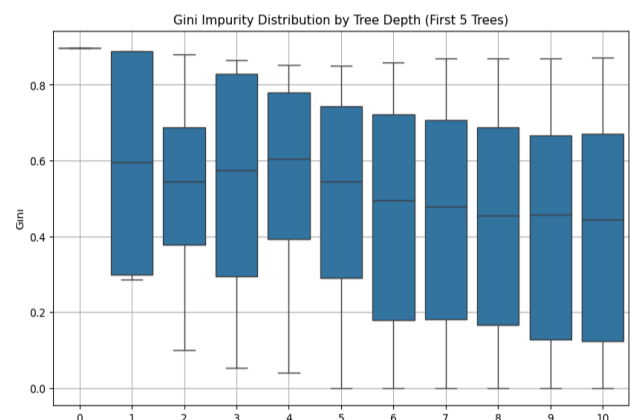
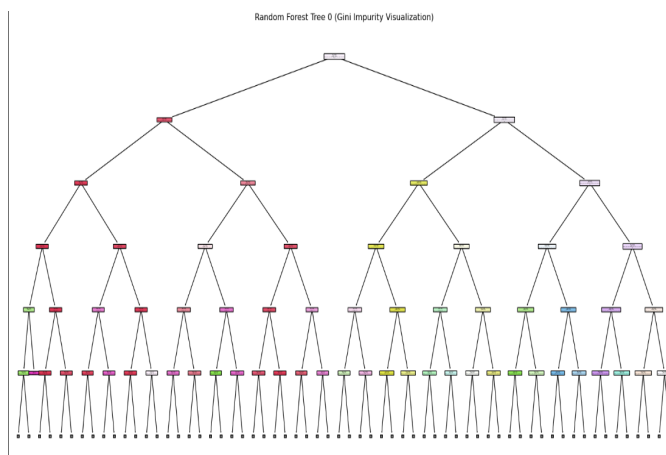
Model A slightly outperforms Model B in terms of macro and weighted averages, with a macro F1-score of 0.53 vs 0.49 and a weighted F1-score of 0.55 vs 0.51 (i.e. Model A handled class balances slightly better and had a more consistent performance across all classes). In terms of precision, Model A again led with a weighted average of 0.55 compared to 0.52 of Model B, suggesting fewer false positives overall. Recall scores were also higher for Model A compared to Model B (weighted avg: 0.56 vs 0.54, respectively), suggesting that Model A had a greater ability to correctly identify true positives, particularly in classes like SeaLake (f1-score: 95%) and Forest (f1-score: 78%). Model B, however, provided a balanced generalization between classes, as its f1-scores are comparable. Model B also did not show stark contrast scores, unlike Model A. Both models struggled with classes such as Highway, River, and HerbaceousVegetation, indicating the need for further model tuning.

## RFC Models Gini impurity values Conclusion

### Gini Impurity Tree Distribution - RFC Model A



### Gini Impurity Tree Distribution - RFC Model B



Model A and Model B differ significantly in their tree structures and impurity profiles. Model A builds deeper trees, up to a maximum depth of forty. A steady decline in Gini impurity is observable as tree depth increases. This indicates a steady decline within the Gini impurity with depth, indicating increasingly pure splits and higher risk of overfitting due to complex tree growth. In Model B, however, tree depth is limited to a maximum depth of 10, resulting in a more compact and interpretable model. Its impurity values, however, are relatively higher across all depths, which indicates underfitting or less effective splits. In summary, Model A demonstrates a capacity to learn complex patterns, though this has been shown to be memorization due to the large degree of overfitting expressed earlier in this section, while Model B favors simplicity and generalization, potentially at the cost of performance.

## Support Vector Machine (SVM)

Support Vector Machine is a supervised learning algorithm that seeks to find the optimal hyperplane that separates classes in the feature space. SVM models are effective in high-dimensional spaces and can model complex, non-linear boundaries using kernel functions, such as the Gaussian (RBF) kernel. They are sensitive to feature scaling and class imbalance. As such, preprocessing (Example: standardization and stratification) is critical. SVMs are widely used for classification tasks where clear class boundaries exist. They perform well with limited, well-structured data.

### SVC Model Results

```
{'svc__C': np.float64(4.821125680265211), 'svc__gamma': np.float64(0.04363677705546425), 'svc__kernel': 'rbf'}
```

Optimized Support Vector Classification Training Accuracy: 52.33%

Optimized Support Vector Classification Testing Accuracy: 51.93%

### SVC Model Overfitting and Accuracy

As configured, the SVC model performed slightly better than random chance (training accuracy: 52.33%, testing accuracy: 51.93%). This suggests that the model is underfitted and that it cannot accurately distinguish between the various classes of data. Enhancing feature engineering or additional model tuning could help boost performance for weaker classes.

### F1-Score, Precision, and Recall

The f1-scores from the Support Vector Machine model revealed mixed performance across classes. It performed very well on SeaLake (f1-score: 0.90) and Forest (f1-score: 0.72), which was reflected by strong precision and recall for these categories. In contrast, several classes showed weak f1-scores, particularly Highway (0.06) and River (0.09), reflecting poor recall and high misclassification rates. These low scores imply that the model struggles to distinguish these classes from others. Other classes, such as HerbaceousVegetation, PermanentCrop, and Residential, fell within the moderate range (f1-scores between 0.37 and 0.44), suggesting inconsistent performance. The macro average f1-score of 0.46 and weighted average of 0.48 confirmed that the model was underperforming, with accuracy hovering around 52%. While the model succeeded at identifying a few dominant classes, it failed to generalize well across all categories.

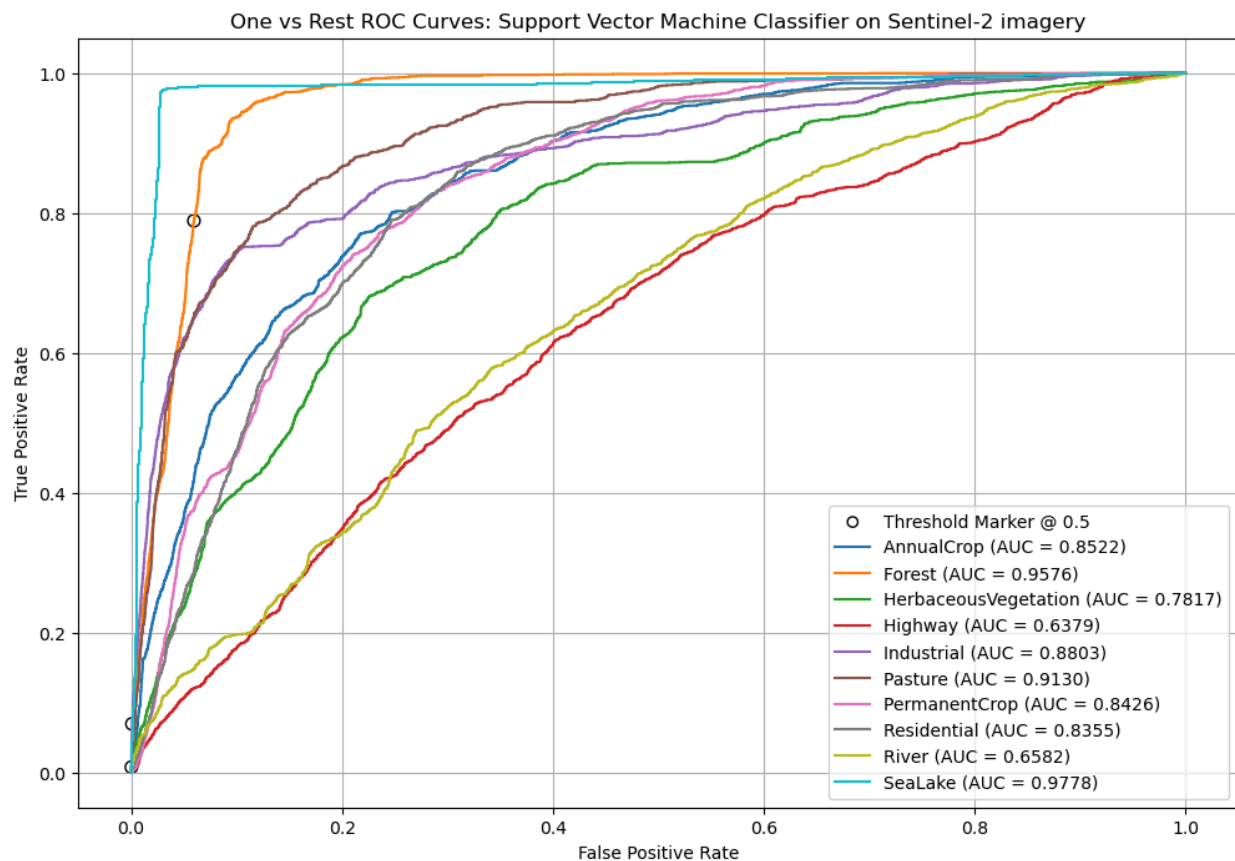
Similarly, precision and recall scores bolster the findings observed in the f1-scores. They demonstrate that the model varies significantly across classes, with SeaLake and Forest showing strong performance (SeaLake achieved a precision of 0.84 and a recall of 0.97, while Forest generated a precision score of 0.60 and a recall of 0.88), indicating the model was highly effective at identifying these categories. In contrast, classes like Highway and River exhibited poor recall (0.03 and 0.05, respectively), meaning the model failed to correctly identify most instances of these types. At 0.49 and 0.50, respectively, the macro average precision and recall

scores reflect the model's uneven performance across classes. This imbalance suggests the model may benefit from class-specific tuning or other remedial actions.

## SVC Model's ROC and AUC Curves

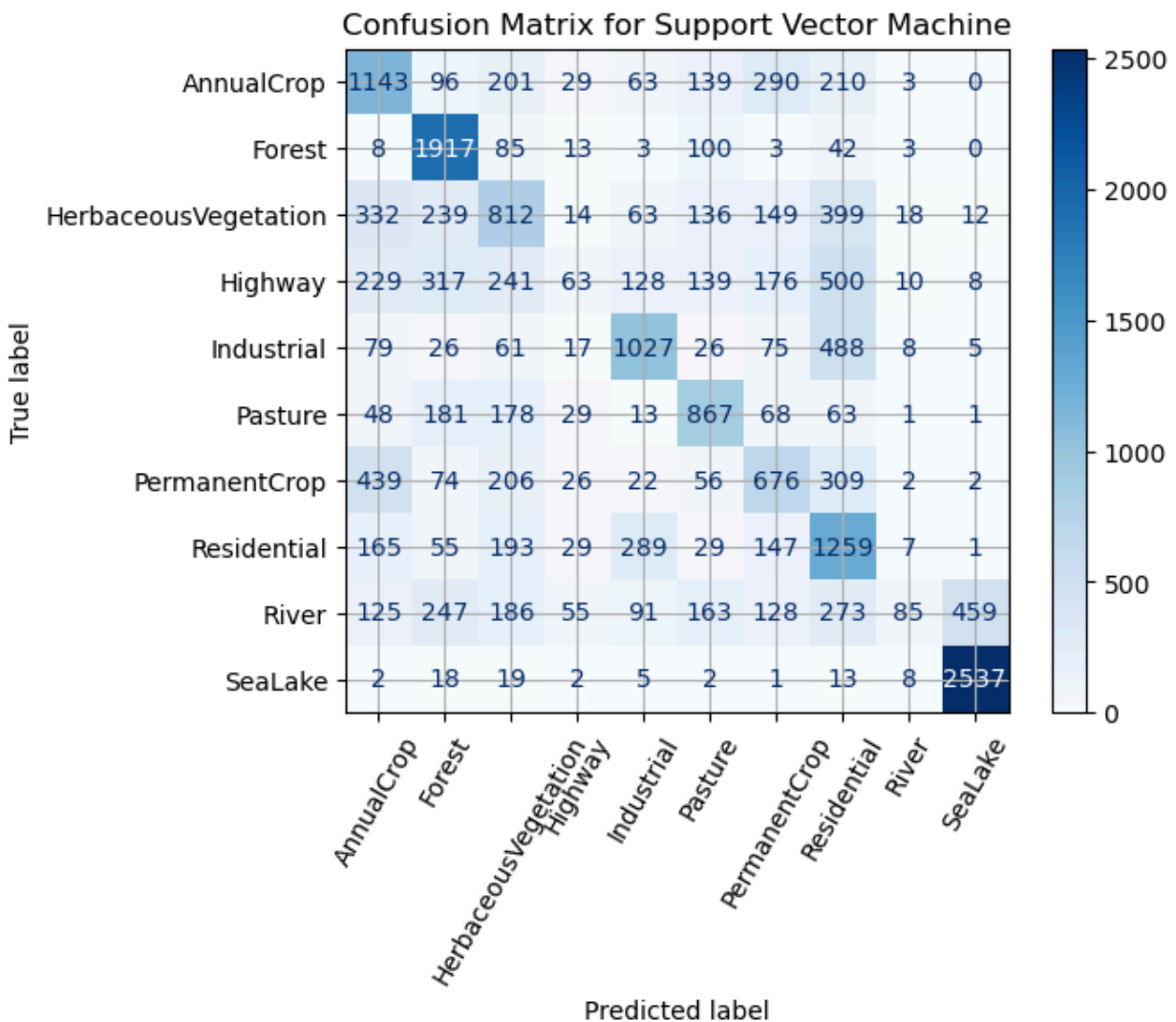
The AUC scores indicate that the model performed well overall, with several classes, such as SeaLake, Forest, and Pasture, showing excellent classification performance (AUC > 0.90). These scores suggest that the model was able to reliably distinguish between these classes due to their distinct features and consistency. Other categories, including AnnualCrop, Industrial, PermanentCrop, and Residential, also showed good performance (AUC ~0.83–0.88).

The model's performance, however, may have suffered due to overlapping characteristics with other classes that made it more challenging to accurately classify. For example, the SVC model struggled with classes like HerbaceousVegetation, Rivers, and Highways, which have notably lower AUC scores (0.78, 0.66, and 0.64, respectively). These results suggest that these classes likely have notable similarities to other classes, or may be underrepresented in the training data. For example, data in the highway class may resemble data in the Industrial or Residential classes, while data in River class may resemble SeaLake or HerbaceousVegetation classes. More than likely, however, this results due to the reason mentioned in the RFC Models ROC and AUC Curves Comparison section (small feature sizes resulting in mixed-class pixels).



## Confusion Matrix

The confusion matrix for the Support Vector Classifier (SVC) model corroborates the previously discussed performance metrics. The SeaLake class exhibited the highest number of true positive predictions (2,537), followed by Forest (1,917) and AnnualCrop (1,143), indicating strong model performance in these categories. Conversely, the River (85) and Highway (63) classes recorded the lowest true positive counts, reflecting the model's limited ability to accurately classify these classes.



## k-Nearest Neighbors (kNN)

k-Nearest Neighbors is a non-parametric, instance-based learning algorithm that classifies samples based on the majority label among their k closest neighbors in the feature space. kNN is simple to implement and interpretable, but its performance depends heavily on selecting an

appropriate number of neighbors (k), distance metric, and feature scaling. It can struggle with high-dimensional data and overlapping class distributions. kNN is best suited for problems where similar instances have similar labels and the feature space is well-behaved.

## kNN Model Results

```
{'knn__weights': 'distance', 'knn__n_neighbors': 11, 'knn__metric': 'euclidean'}
```

Optimized k-Nearest Neighbor Classification Training Accuracy: 99.91%

Optimized k-Nearest Neighbor Classification Testing Accuracy: 54.83%

## kNN Model Overfitting

The K-Nearest Neighbors (kNN) model showed signs of significant overfitting, achieving a near-perfect training accuracy of 99.91% with a much lower test accuracy of 54.83%. This gap suggests the model memorized the training data rather than learning generalizable patterns. While the overall accuracy is slightly better than random chance, the performance remains limited in its ability to separate all land cover classes reliably. The model's distance-based approach likely struggled with overlapping feature distributions.

## F1-Score

F1 scores for the kNN model show high variance across different classes. SeaLake (f1-score: 0.95) and Forest (f1-score: 0.77) stand out as well-classified categories, with strong recall and precision. SeaLake, for example, achieved both high recall (0.96) and precision (0.94), meaning the model correctly identified most SeaLake samples and made very few incorrect SeaLake predictions.

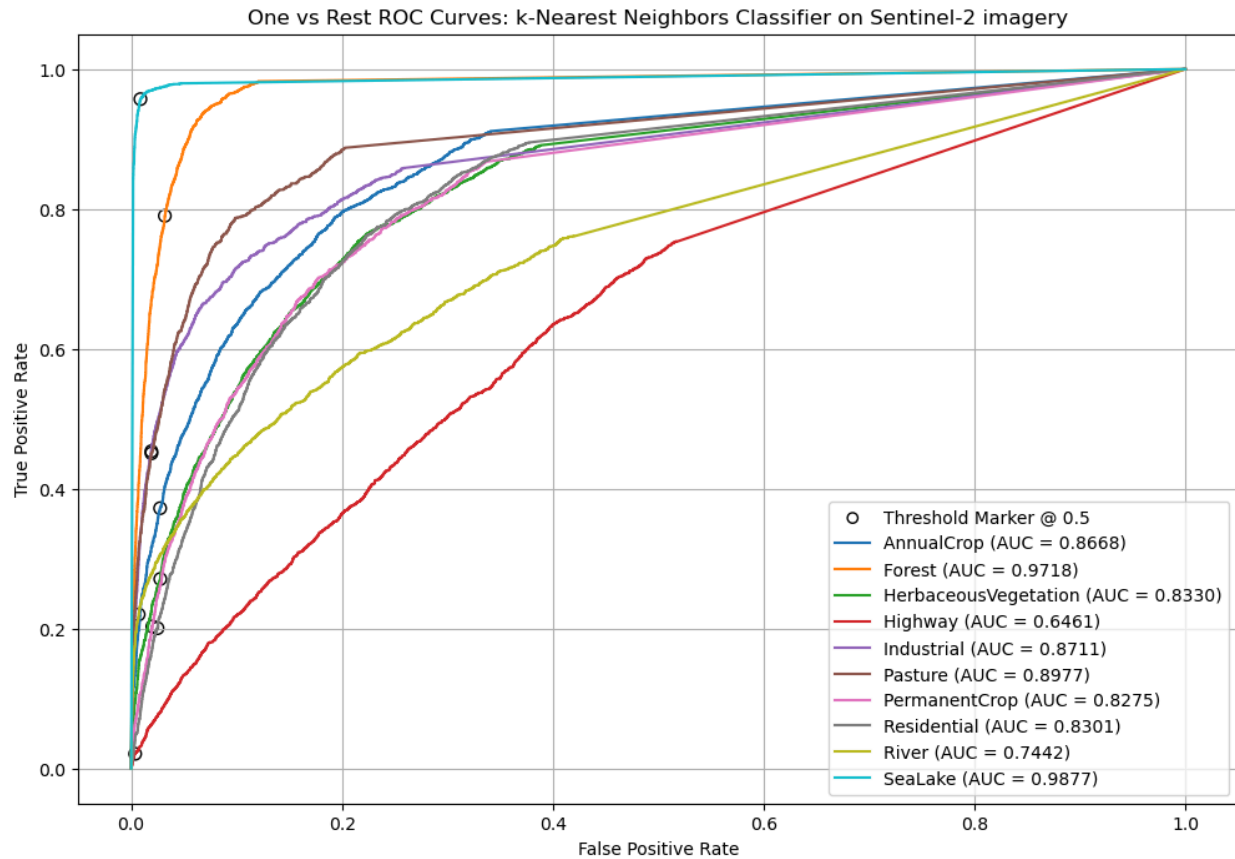
In contrast, classes like Highway (f1-score: 0.14) and River (f1-score: 0.38) performed poorly. Highway had a very low recall (0.10), meaning many true Highway samples were missed, and a low precision (0.21), indicating many false positives. River also suffered from poor recall (0.30), which implies that the model frequently misclassified River pixels as other categories.

Other classes, such as HerbaceousVegetation (f1-score: 0.45), Residential (0.42), and PermanentCrop (0.40), had middling precision and recall, indicating inconsistent performance. The macro average f1-score of 0.52 and weighted average of 0.53 reflect this inconsistency and confirm that the model fails to generalize well across most classes, despite strong results in a few.

## kNN Model ROC and AUC Values and Curve Results

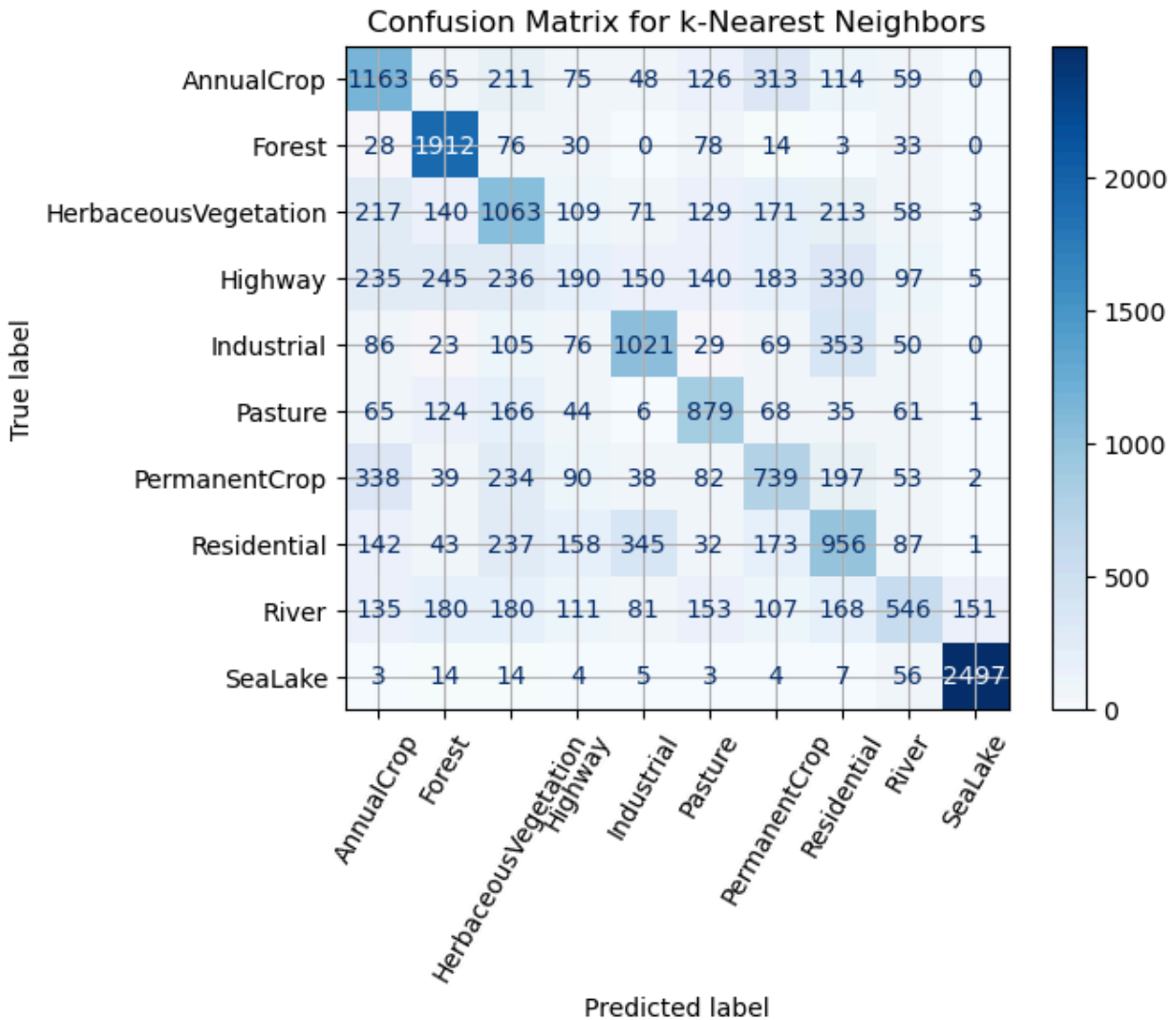
The AUC scores generated by the kNN model highlight its uneven classification ability. Classes such as SeaLake (AUC: 0.9877), Forest (0.9718), and Pasture (0.8977) had excellent separability, suggesting distinct visual or spectral characteristics. Other well-performing categories include Industrial (0.8711), AnnualCrop (0.8668), and PermanentCrop (0.8275), where the model could distinguish features with moderate success.

In contrast, classes like Highway (AUC: 0.6461) and River (0.7442) showed lower AUC values, implying frequent misclassification and overlap with other categories. These weak results may stem from visual similarities between River and SeaLake or Highway and Industrial/Residential zones.



## Confusion Matrix

The confusion matrix supports the above findings. SeaLake achieved the highest true positives (2,497), followed by Forest (1,912) and AnnualCrop (1,163), reinforcing the model's strength with these categories. On the other hand, River (546) and Highway (190) had lower correct prediction counts, suggesting difficulty in isolating these classes. The dispersion of misclassified values across similar classes also reflects the model's sensitivity to unscaled distances and lack of generalization.



## Conclusion

### Model Performance Overview

The three classifiers: Random Forest (RFC), Support Vector Machine (SVC), and k-Nearest Neighbors (kNN), were evaluated on the EuroSAT satellite imagery dataset using accuracy, precision, recall, f1-score, ROC AUC, and confusion matrices. Each model was optimized via hyperparameter tuning and assessed on a stratified test set to ensure fair comparison.

### Accuracy and Overfitting

- RFC Model A achieved the highest training accuracy (86.85%) but suffered from overfitting, with a test accuracy of 56.32%. RFC Model B, with restricted tree depth, generalized better (training: 55.68%, test: 53.55%).

- SVC showed balanced but lower accuracy (training: 52.33%, test: 51.93%), indicating underfitting and difficulty in separating all classes.
- kNN had near-perfect training accuracy (99.91%) but lower test accuracy (54.83%), a classic sign of overfitting due to memorization of training data.

## Precision, Recall, and F1-Score

- RFC models performed best on dominant classes like SeaLake and Forest, but struggled with minority classes such as Highway and River.
- SVC and kNN showed similar trends, with high scores for prevalent classes and poor performance for underrepresented or visually similar classes.
- Macro and weighted averages for f1-score and precision were highest for RFC Model A, but the gap between training and test scores highlights overfitting.

## ROC, AUC, and Confusion Matrix

- All models achieved high ROC AUC for easily separable classes (SeaLake, Forest), but lower scores for classes with overlapping features (Highway, River).
- Confusion matrices revealed frequent misclassifications among spectrally similar land cover types and land cover types with similar feature sizes, especially for Highway and River.

## Generalization and Model Suitability

- RFC Model B offers better generalization and interpretability due to limited tree depth, making it more reliable for unseen data.
- SVC is robust to feature scaling and performed consistently, but may require further tuning or additional features to improve class separation.
- kNN is highly sensitive to class imbalance and feature overlap, leading to overfitting and poor generalization.

## Conclusion

The results of this study differed from the conclusions generated by the authors assessed during our literature review. Phan Thanh Noi and Martin Kappas determined that a SVC model would be the most robust model for analyzing land cover in their study (2018). However, the RFC Model B emerged as the best overall model in this analysis. It struck an effective balance between accuracy and generalization, though further tuning should be performed to achieve better accuracy scores.

Findings from the analysis highlight the importance of feature scaling and stratification, particularly for models like SVC and kNN. In contrast, RFC Model B demonstrated robustness against multicollinearity and class imbalance, though it requires careful tuning to prevent overfitting. Looking ahead, future work could focus on exploring resampling techniques, like

SMOTE (Synthetic Minority Over-sampling Technique), to generate synthetic data points for rare classes; using a soft voting ensemble model to determine the best metrics among existing classification models; implementing advanced feature engineering techniques; Using boosting algorithms, such as XGBoost and LightGBM, which are often superior to standard Gradient Boosted Trees, and inherently handle class imbalance; Continued optimization of the SVC model by exploring a wider range of kernel types; or leveraging extra boost random forest classification and deep learning approaches to enhance classification performance, especially for minority classes, those with spectrally similar features, and high visual similarity.

# References

- Gotam, D., & Debnath, A. (n.d.). EuroSAT dataset [Data set]. Kaggle. Retrieved July 12, 2025, from <https://www.kaggle.com/datasets/apollo2506/eurosat-dataset>.
- Intel Corporation. (2025). Intel Extension for Scikit-learn (scikit-learn-intelex) (Version 2025.7.0) [Computer software]. Python Package Index. <https://pypi.org/project/scikit-learn-intelex/>.
- Phan Thanh Noi, & Kappas, M. (2018). Comparison of Random Forest, k-Nearest Neighbor, and Support Vector Machine classifiers for land cover classification using Sentinel-2 imagery. *Sensors*, 18(1), 18. <https://doi.org/10.3390/s18010018>.
- Sinergise. (n.d.). Sentinel-2 bands | Sentinel Hub custom scripts. Sentinel Hub. Retrieved July 12, 2025, from <https://custom-scripts.sentinel-hub.com/custom-scripts/sentinel-2/bands/>.